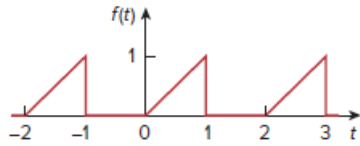


Problem

Obtain the Fourier coefficients of the function in Fig. 17.7 using *PSpice*.

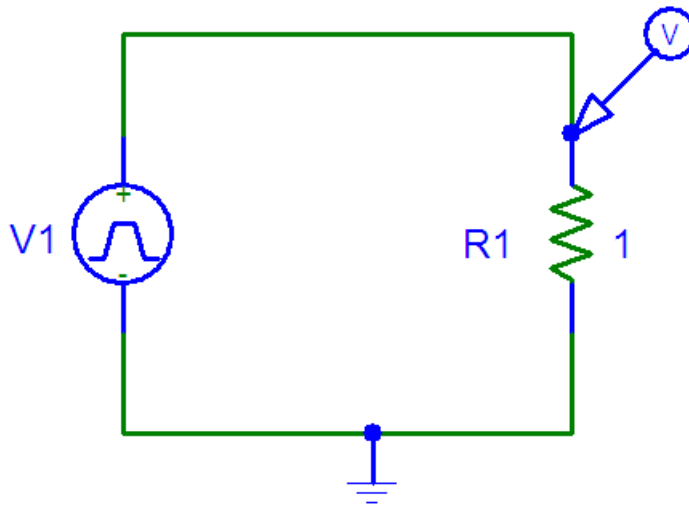
Figure 17.7:



Step-by-step solution

Step 1 of 9

Step 1: Using PSpice to find the Fourier components consisting of four steps.

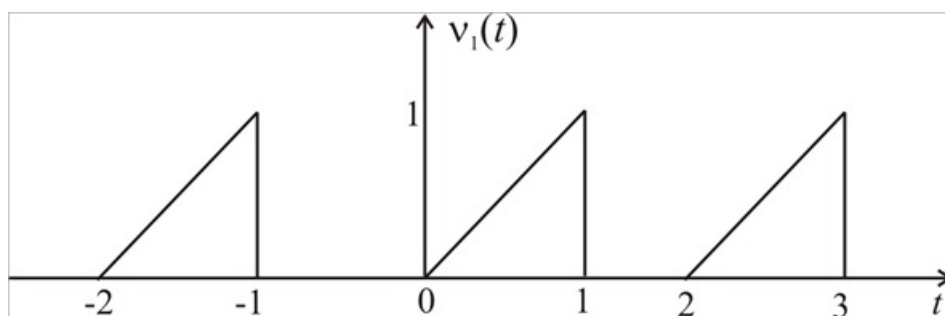


Step 2 of 9

The schematic for obtaining the Fourier coefficients

Step 3 of 9

The given input signal is,



Step 4 of 9

Step 2: We enter the attributes of the voltage source VPULSE by double clicking on the voltage source and enter the value as

$$V_1 = 0$$

$$V_2 = 1$$

$$TD = 0$$

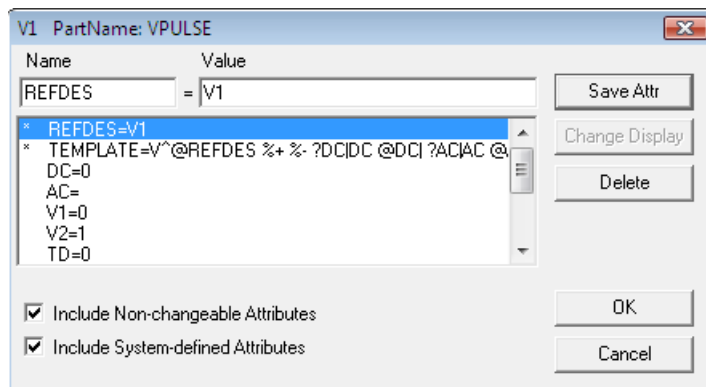
$$TR = 1$$

$$TF = 1\mu s$$

$$PW = 0.01$$

$$PER = 2$$

Step 5 of 9



Step 6 of 9

Step 3: From the given signal it is evident that $T = 2s$

$$f_0 = \frac{1}{T}$$

$$f_0 = \frac{1}{2}$$

$$f_0 = 0.5 \text{ Hz}$$

Step 7 of 9

In the menu bar

Click **analysis** → **setup**

In this setup disable the bias point and enable the **transient** and click on transient tool box and enter the parameter as

Final time as $5T = 10s$

Print setup as $0.05s$

Step ceiling as $10ms$

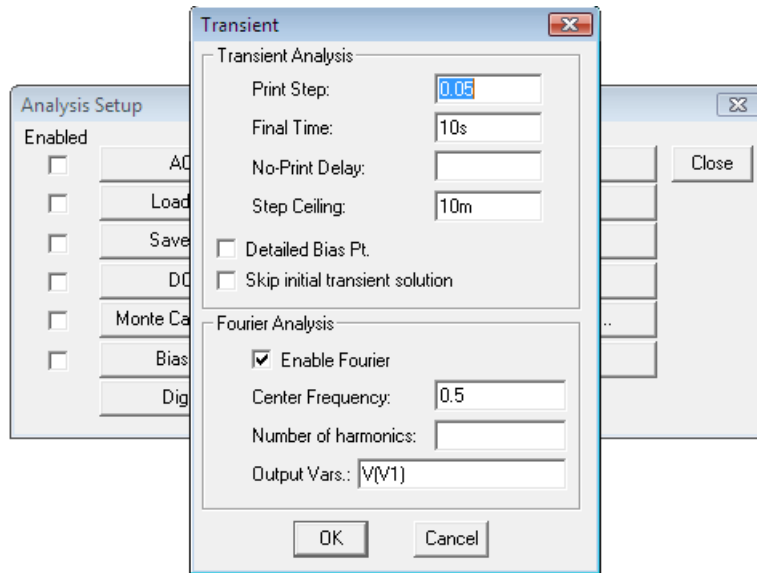
In the **Fourier analysis**

Tick **Enable Fourier**

Center frequency as 0.5 Hz and the

Output VARS as $V(V1)$

Step 8 of 9



Step 9 of 9

Step 4: Then save and click on **simulate**, when PSpice is run, the output file contains the result as

FOURIER COMPONENTS OF TRANSIENT RESPONSE V(V_V1)

DC COMPONENT = 2.598758E-01

HARMONIC FREQUENCY FOURIER NORMALIZED PHASE NORMALIZED

NO (HZ) COMPONENT COMPONENT (DEG) PHASE (DEG)

1 5.000E-01 3.879E-01 1.000E+00 -3.500E+01 0.000E+00

2 1.000E+00 1.593E-01 4.106E-01 1.729E+02 2.429E+02

3 1.500E+00 1.126E-01 2.904E-01 -2.203E+01 8.296E+01

4 2.000E+00 7.977E-02 2.057E-01 1.658E+02 3.058E+02

5 2.500E+00 6.685E-02 1.724E-01 -2.446E+01 1.505E+02

6 3.000E+00 5.334E-02 1.375E-01 1.587E+02 3.686E+02

7 3.500E+00 4.773E-02 1.231E-01 -2.947E+01 2.155E+02

8 4.000E+00 4.018E-02 1.036E-01 1.515E+02 4.315E+02

9 4.500E+00 3.723E-02 9.599E-02 -3.537E+01 2.796E+02